



Improving Driving Information interfaceability among Vehicles through an Online Dashboard

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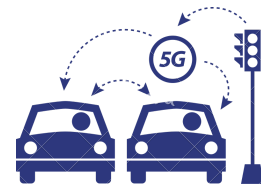
Introduction to V2X Communication

V2X (Vehicle-to-Everything) communication is a cutting-edge technology revolutionizing transportation by enabling real-time, two-way information exchange among vehicles, infrastructure, pedestrians, and the environment. It furthers the efforts in Inter-Vehicle Communication (IVC) thereby uplifting Intelligent Transport Systems (ITS).



V2V Communication

Enables Vehicle to Communicate directly with each other, enhancing safety.



V2I Communication

Facilitates interaction between vehicles and infrastructure



V2P Communication

Allows vehicles to communicate with pedestrians, increasing safety measures.



V2G Communication

Connects vehicles to the grid, enabling energy management and usage optimization



V2N Communication

Links vehicles to the network, providing access to the data and services



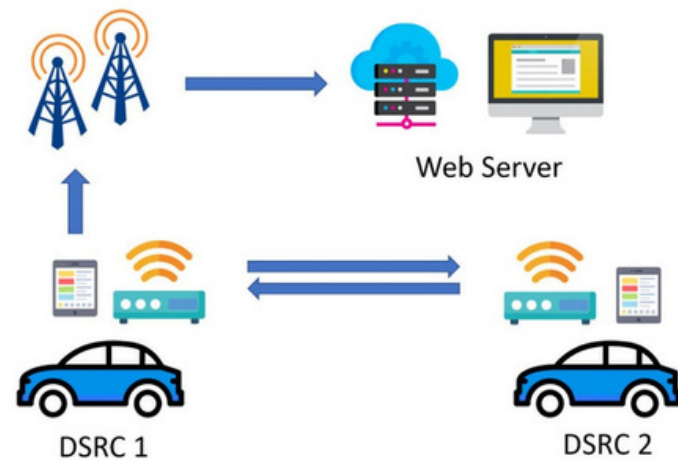
V2D Communication

Enables vehicles to interact with various devices, enhancing functionality.

Background Research: Advancements in V2X

Technologies used

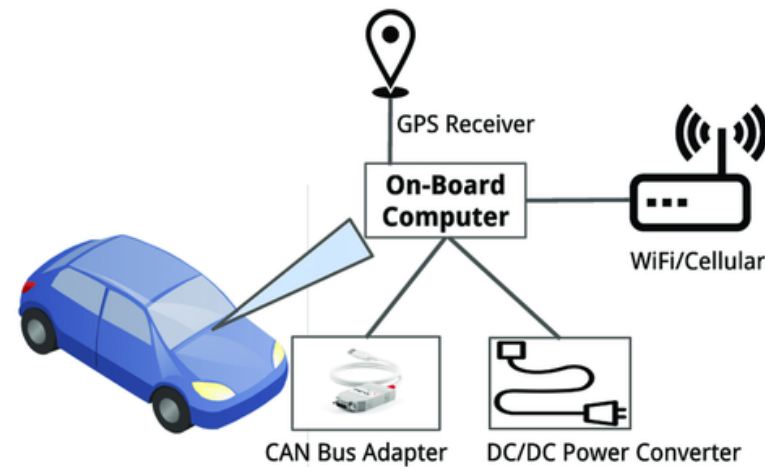
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Dedicated Short-range Communication (DSRC)

DSRC is a key technology enabling real-time vehicle communication, enhancing safety and traffic management.

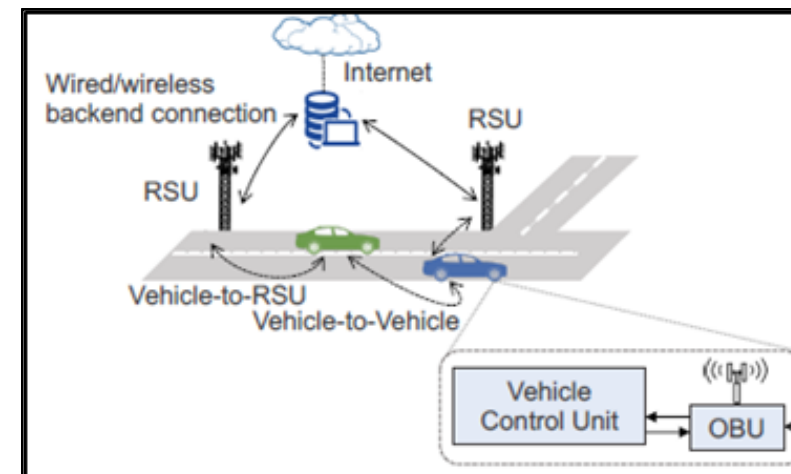
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On-Board Units (OBUs)

OBUs are installed in vehicles facilitate communication with infrastructure and other vehicles, crucial for V2X functionality

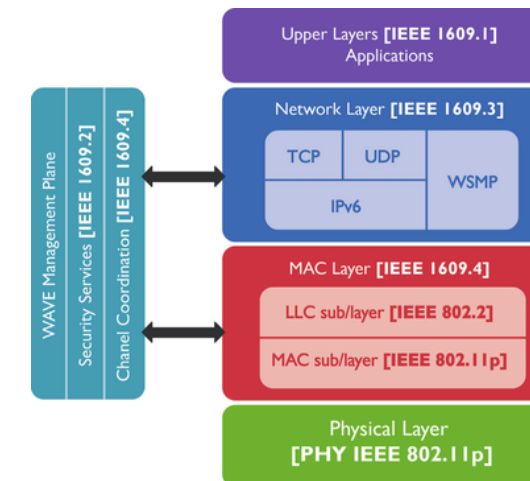
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Roadside Units

RSUs serve as communication hubs at intersections, enabling seamless data exchange between vehicles and traffic systems. [10]

4



IEEE 802.11p / WAVE

Wireless access in vehicular environments (WAVE) performs at 5.9 GHz and generally works on short range communication and IEEE 802.11p defines its protocols. [6]

5



Cellular V2X (C-V2X)

C-V2X uses cellular networks like 4G LTE, 5G. Much greater range compared to the ones from IEEE 802.11p

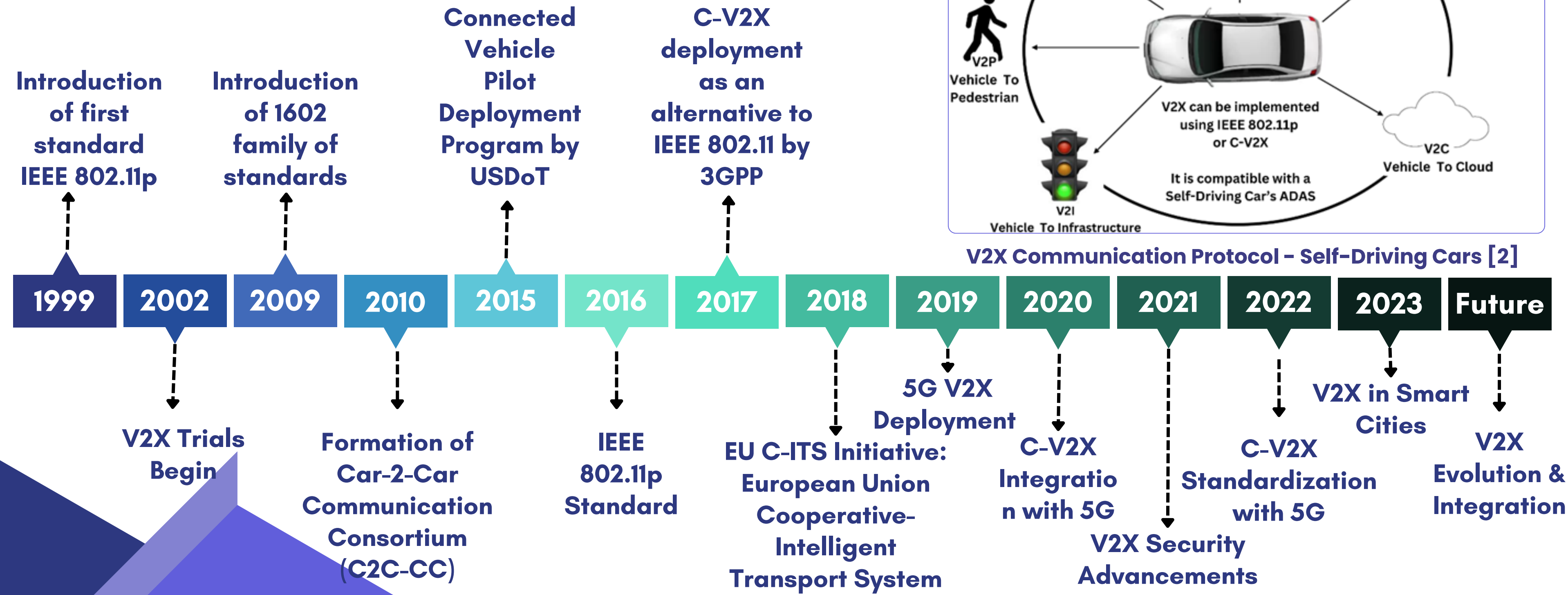
Background Research: Advancements in V2X

Literature Review

- 1 LoRa-Based V2X Communication** An IVC system integrates OBU and RSU technologies in V2X, using RSSI and distance for energy-efficient data transfer, with LPWAN ensuring secure vehicle communication. [3]
- 2 Blockchain application for V2V** Dashcam video data is managed via multi-signature verification, GPS-based distance, a decentralized Oracle, HDFS, and Hyperledger Fabric.[4]
- 3 GCSS Based Scheme for V2V** Graph Coloring-Based Spectrum Sharing (GCSS) maximizes system capacity and spectrum usage by optimizing cellular links, V2V clusters, and system capacity through graph coloring. [5]
- 4 WAVE for Traffic Management System** Proposes a V2X-based Traffic Management System using OBUs, RSUs, IEEE 802.11p, and DSRC to enhance traffic flow dynamics. [6]
- 5 Driver Drowsiness Detection with Eye Aspect Ratio** Uses Eye Aspect Ratio (EAR) from facial landmarks for a 3-part system in driver drowsiness detection, alerting the driver and nearby law enforcement to prevent accidents. [7]
- 6 LiFi for enabling V2P & V2I** Develops Optical V2V communication for automotive environments using LiFi and Visible Light Communication (VLC) systems.[8]
- 7 Traffic Management for Emergency Vehicles** A system optimizing vehicle proximity and packet delivery ratio to minimize end-to-end delay, ensuring reliable emergency pathways in heavy traffic. [9]

Evolution of V2X Technology

A timeline showcasing significant milestones in V2X advancements



Our Proposed Work

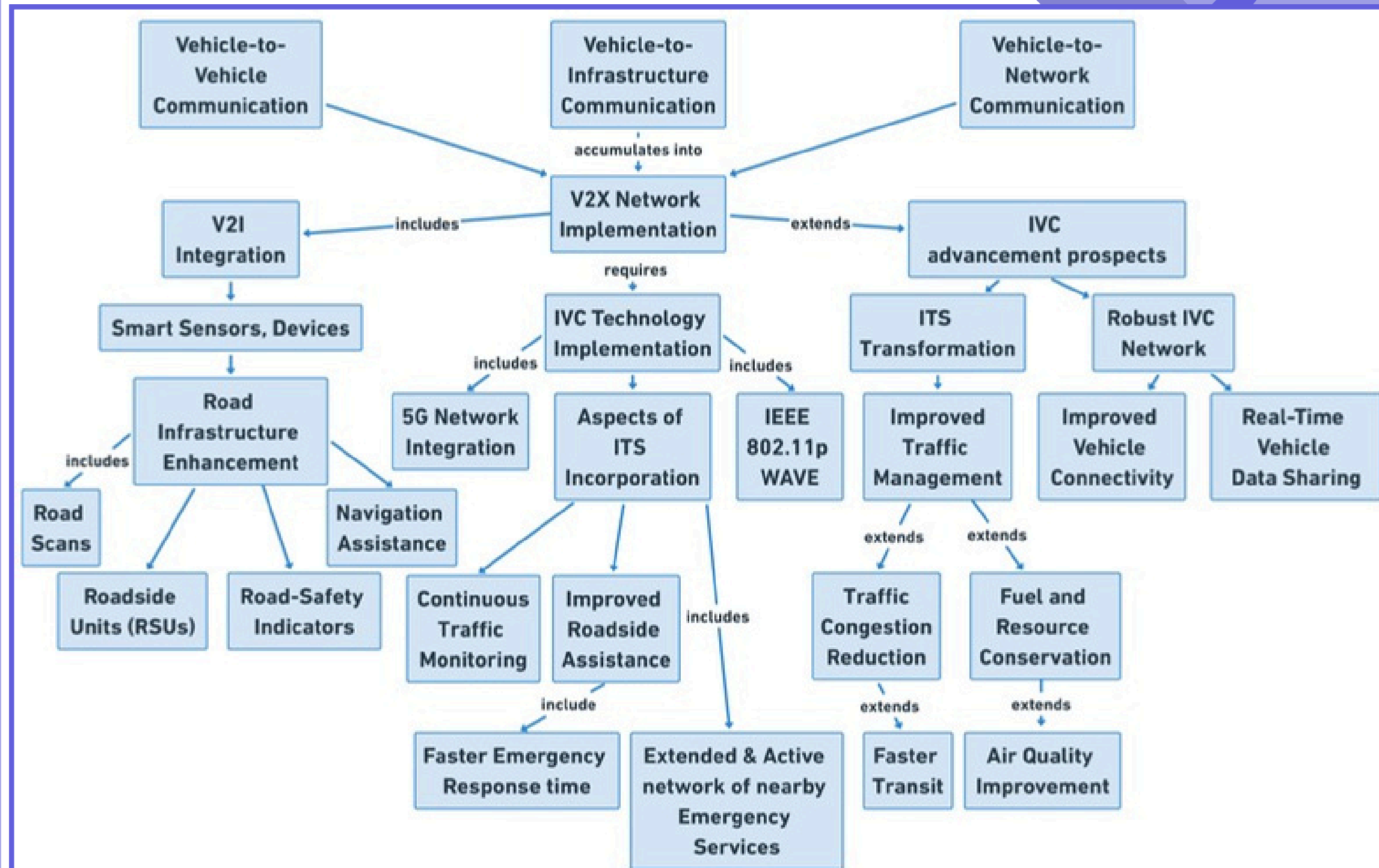
01

- The proposed work aims to create an online dashboard for connected vehicles, offering real-time monitoring of driving situations and emergency assistance.
- Users can access data on – traffic, weather, road conditions, and emergency warnings.
- It also facilitates cooperative roadside assistance for issues like – car breakdowns and locating nearby services like garages, hospitals, and gas stations.

02

The dashboard attempts to –

- Improve traffic management,
- Enable prompt and efficient emergency responses, and
- Increase driver safety by presenting this information in a centralized and easily accessible manner.
- The dashboard will make use of the infrastructure and V2V communication technologies already in place to guarantee dependable operation and a smooth integration.



Flowchart of Proposed Work

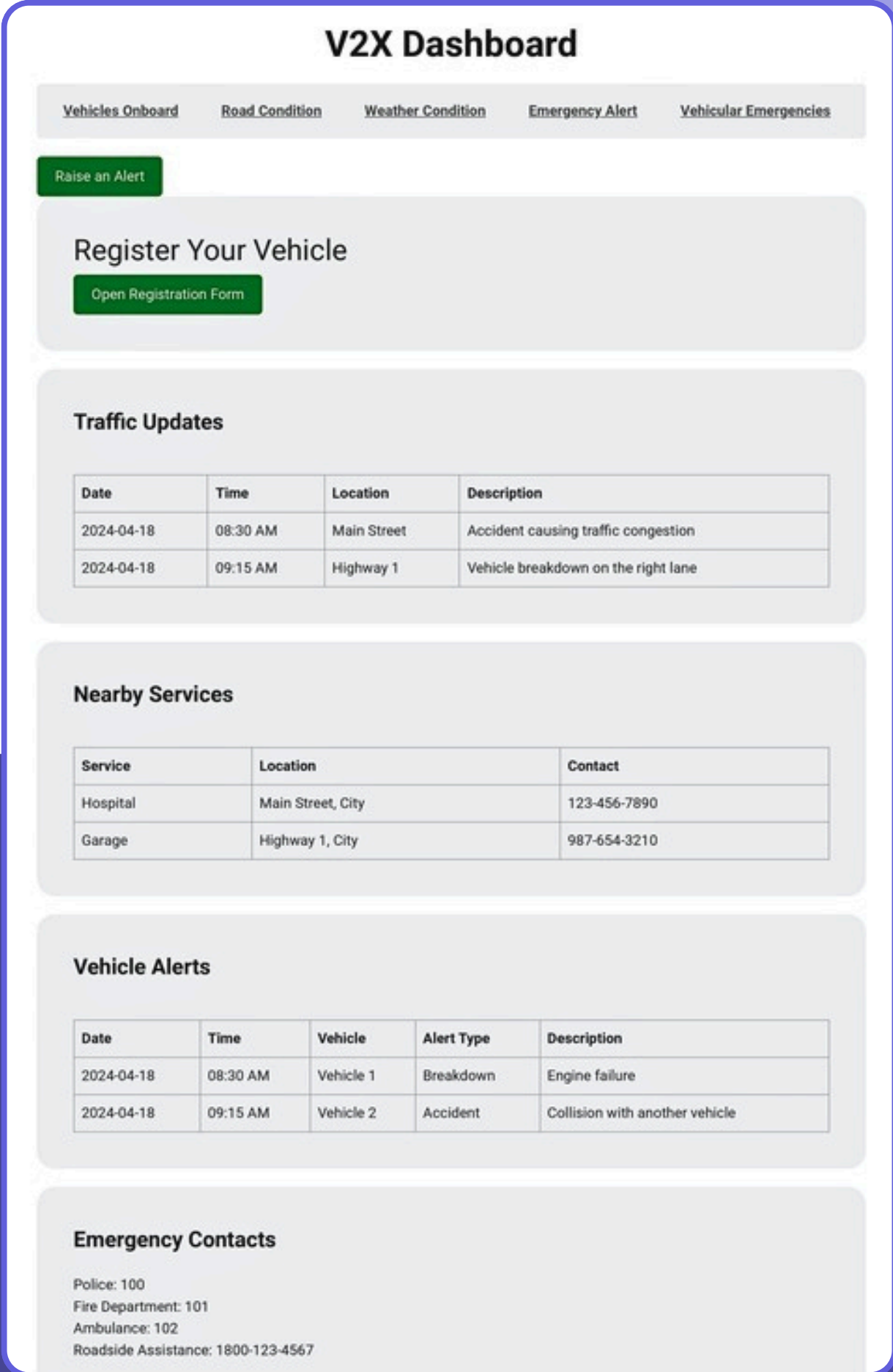
Methodology

01

- **Enhanced User Experience:** The dashboard design aims to improve user experience and safety.
- **Clear Display of Information:** Important information is shown prominently; less important details are hidden.
- **Modular Design:** The homepage breaks into multiple pages for easy navigation.
- **Expandable and Adaptable:** New features can be added by creating or updating pages.

02

- **V2X Dashboard Homepage:** Navigation links to “Vehicles Onboard,” “Road-condition,” “Weather-condition,” “Emergency Alerts,” and “Vehicular Emergencies.”
- **Alert & Registration Forms:** For distress alerts and vehicle network registration.
- **Traffic & Emergency Info:** Real-time traffic updates, nearby emergency services, and vehicle alerts.
- **Emergency Contacts:** Key contact information for emergencies.



V2X Dashboard Homepage

Methodology

V2X Dashboard

Vehicles Onboard

Road Condition

Weather Condition

Emergency Alert

Vehicular Emergencies

Raise an Alert

Register Your Vehicle

Open Registration Form

The V2X Dashboard is a web app for V2X communication, offering real-time updates on traffic, weather, and emergency alerts to improve driving safety and provide access to emergency services and cooperative driving features.

Road Condition

Kolkata

Mumbai

Delhi

Chennai

Hyderabad

Jaipur

Chandigarh

Guwahati

Bangalore

Raipur

Jamshedpur

Bhopal

Kanpur

Pune

Nagpur

Patna

Ahmedabad

Lucknow

Noida

Nasik

Surat

Kochi

Coimbatore

Durgapur

Received Updates

Road-condition updates received from vehicles on Network

Date	Time	Vehicle IP Address	Vehicle Name	Vehicle Sensor Type	Message Received
2024-04-16	12:00 PM	192.168.1.1	Vehicle 1	Sensor Type 1	Message 1

See Road Condition

City:

Pune

Road Type:

Expressway

Road Name:

Expressway 1

Show Road Condition

Road Condition for Expressway 1

of RSA Units: 60

of V2X Connected Infrastructure: 110

% of Road Mapped under V2X Network: 65

Driving Condition: 9/10

Vehicles Onboard

View more

Vehicle Statistics

No. of Vehicles: 2

Passenger Vehicles: 1

Commercial Vehicles: 1

Vehicles with V2V Enabled: 0

Vehicles with Direct V2X Enabled: 0

Vehicle Data

All Vehicles

ABC123 1 192.168.1.1

XYZ456 2 192.168.1.2

New Vehicles Onboarded

PQR789 3

LMN012 4

Vehicles Deregistered

XYZ456 2

Register Your Vehicle

Open Registration Form

The Road Conditions Dashboard provides real-time information about road conditions in various cities, aiming to improve V2X connectivity and V2V communication. Users can select a city, road type, and road name to view detailed updates.

The Vehicles Onboard Dashboard enhances V2X connectivity and V2V communication by providing real-time vehicle information and statistics, including tracking new vehicles and managing deregistered ones.

Methodology

Weather Condition

Kolkata

Mumbai

Delhi

Chennai

Hyderabad

Jaipur

Chandigarh

Guwahati

Bangalore

Raipur

Jamshedpur

Bhopal

Kanpur

Pune

Nagpur

Patna

Ahmedabad

Lucknow

Noida

Nasik

Surat

Kochi

Coimbatore

Durgapur

Received Updates

Weather updates received from vehicles on Network

Date	Time	Vehicle IP Address	Vehicle Name	Vehicle Sensor Type	Message Received
2024-04-16	12:00 PM	192.168.1.1	Vehicle 1	Sensor Type 1	Message 1

The Weather Conditions Dashboard provides real-time information about weather conditions in various cities, improving V2X connectivity and V2V communication. Users can select a city to view detailed updates.

Emergency Alerts and Services

Accident

Natural Hazard

Health Hazard

Industrial Hazard

Political Event

Road Blockage

Cooperative Roadside Assistance

Emergency Services

Emergency Services

Hospitals

Garages

Petrol Pumps

Stations

Food Places

Medical Stores

Garages

Store Name: Example Store

Store Address: 123 Example St

Contact: 123-456-7890

Number of Outlets: 5

Locate nearest Garage

Received Alerts

Received updates from vehicles regarding emergency alerts

Vehicle Numberplate	Vehicle IP Address	Vehicle Name	Date	Time	Hazard Type	Message Received
ABC123	192.168.1.1	Vehicle 1	2024-04-16	12:00 PM	Accident	Message 1

The Emergency Alerts and Services Dashboard provides real-time information about emergency situations and services in different locations, enhancing road safety and emergency response. Users can access alerts, detailed emergency information, and service locators.

Vehicular Emergencies

Raise an Alert

Raise an Alert

Location

Enter location

Driver Name

Enter driver name

Vehicle Name

Enter vehicle name

Vehicle Number-Plate

Enter vehicle number-plate

Type of Emergency:

Vehicle Breakdown

Emergency Description

Enter emergency description

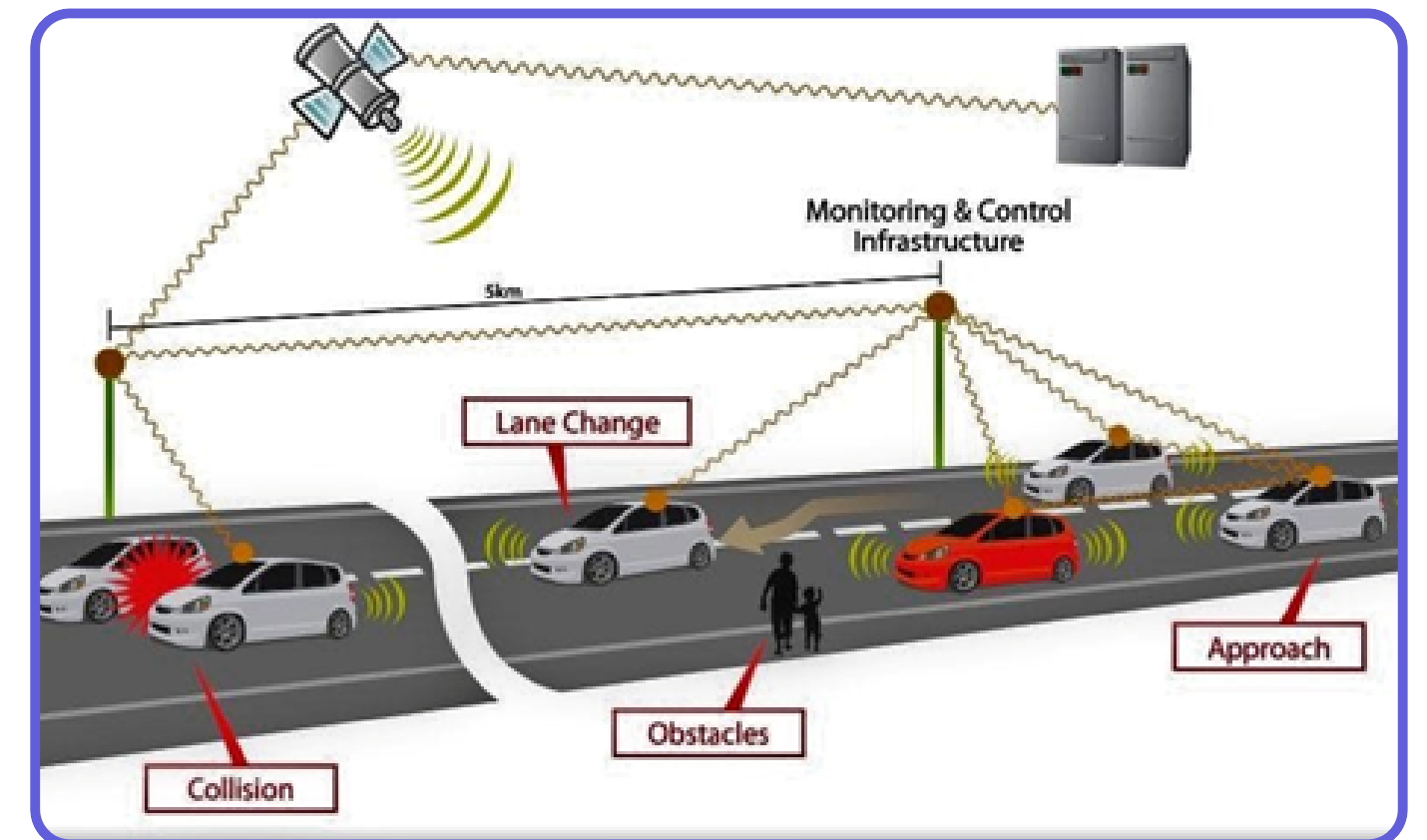
Submit

Date	Time	Location	Vehicle Numberplate	Vehicle IP Address	Vehicle Name	Driver Name	Help Receiver/Offeor	Emergency Type	Emergency Detail	Resolution
2022-01-01	10:00 AM	Main Street, Mumbai	ABC123	192.168.1.1	Maruti Suzuki Swift	Rahul Shah	Receiver	Breakdown	Engine failure	Towed to the nearest garage

The Vehicular Emergencies Dashboard handles emergency situations efficiently, serving as a platform for raising alerts and maintaining records within the V2X network. It enhances safety standards and improves emergency response in vehicular environments.

Conclusion

- ➔ The development of an online V2V Dashboard is a significant step forward in improving driver safety and traffic management.
- ➔ The project utilizes V2X communication technologies to establish a centralized platform for real-time monitoring and emergency assistance.
- ➔ By offering drivers essential information like traffic updates and weather conditions, the dashboard boosts awareness and facilitates proactive decision-making on the road.
- ➔ Moreover, the project addresses the limitations of current ADAS systems by merging V2V and V2X communication capabilities. This integration promotes cooperative driving, enabling vehicles to exchange data to prevent accidents and enhance traffic flow.
- ➔ The project utilizes V2X communication technologies to establish a centralized platform for real-time monitoring and emergency assistance.
- ➔ Looking ahead, the V2X Dashboard project holds vast potential for further growth and expansion. Future endeavors may concentrate on refining features, enhancing V2X connectivity, and broadening its outreach to more vehicles and infrastructure.



Information exchange – V2V Communication [1]



Thank You

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